A course advisory dataset for IIT Gandhinagar M.Tech students

CS 203 Assignment 1, Project 4

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1 Overview

We built a dataset for a small language model that answers three kinds of M.Tech advising questions: what is still left for my degree, can I register for a course next term, and which electives should I take. The scope is four programmes in the CS and EE departments, MTech-CSE, MTech-AI, MTech-EE and MTech-ICDT, each with a thesis and a coursework track [E1]. A rule engine built from the institute's own documents computes every gold answer, so each label points back to a source page. A small local LLM (Qwen2.5-1.5B-Instruct) paraphrases questions and justifications, and two humans audit those paraphrases. The curriculum has 104 courses and 77 rules with provenance, and the engine generates 180 synthetic students [E2, E3, E4].

600	1526	80	76	19
advisory pairs, 200 per intent [E5]	verified augmented pairs [E6]	doubly labelled audit items [E7]	frozen gold test items [E8]	real form questions (shifted set) [E9]

Every number here is copied from a repository file; Appendix A lists them as E1, E2, …, cited in square brackets.

2 Part A: Collection

We collected the institute's rules and course lists, kept every raw file untouched, and built a cleaned curriculum in which every record says where it came from. The main finding is that no source states an enforceable postgraduate prerequisite.

104	77	22	3083	0
courses [E2]	degree rules, 20 of them EE [E3]	course plan files [E15]	metadata rows, one per sample [E11]	enforceable PG prerequisites [E19]

Sources and provenance. The raw layer holds eight academic advisories, the revised CSE M.Tech course structure 2025-26, the Sem-I 2026-27 timetable and four pages of the EE M.Tech site, each with a sha256 [E10]. Every course and rule record carries its source document with page, row or section and its own hash, and every rule stores a verbatim quote that the build checks against the page text. The metadata file has 3083 rows (578 raw, 2505 processed): source files, pages, timetable rows, plans, catalog records, every rule with a stated value (76 of 77), 180 students, 600 pairs, 1526 augmented children and 19 form responses [E11, E12].

Course plans. For the 50 CS and ES codes, 15 plans downloaded, 12 links need an IITGN login (not bypassed) and 23 codes have no link [E13]. For the 25 EE codes the split is 7, 4 and 14 [E14], so 22 plan files were saved [E15].

Catalog and rules. The raw catalog keeps 136 records with their defects (duplicates, the "Algrithms" typo, float credits) [E16]. Cleaning turns them into 104 courses (67 basket, 15 open elective, 6 stream core, 2 compulsory, 6 requirement, 8 UG background) and logs all 78 changes [E17, E18]. One rule, maximum credits per semester, is marked not stated because no source gives it [E3].

Prerequisites. 103 of 104 courses state no prerequisite; the other, CS 613, has a manual entry with an empty list, because its plan says "Prerequisite (Optional)" with UG courses "or equivalent" (proposed by the pipeline assistant, confirmed by hem436) [E19, E20]. So the engine has two tiers: hard prerequisites (checked, empty in practice) and recommended background, which 3 courses carry [E21]. The prerequisite graph has 104 nodes, 9 edges (all recommended), no missing codes and no cycles [E22]. Decision D4 stores UG codes from plans as recommended background, and D5 (hem436) says they never warn an M.Tech student [E23].

Students, consent and data flow. 180 synthetic students come from seed 203: CSE 64, EE 51, AI 41, ICDT

24; 92 thesis and 88 coursework; 1422 transcript entries with 0 rule violations [E4, E24]. The documents are public regulations used for non-commercial coursework. Form respondents tick a required consent box, no names or emails are collected, and contact details in free text are removed. DATA_FLOW.md draws the flow from sources to Label Studio.

3 Part B: Validation

A suite of 25 checks runs on the raw catalog, the processed data and a deliberately corrupted copy [E25]. It finds the raw defects, passes after cleaning and catches every injected fault. Human review still found two engine bugs that the checks could not see.

25	7 of 10	24 + 1	15 of 15	2
checks: Pandera, networkx, rapidfuzz, rule engine [E25]	raw-catalog checks fail [E26]	processed: pass + one warning [E27]	injected faults caught [E29]	gold bugs found by people [E33, E34]

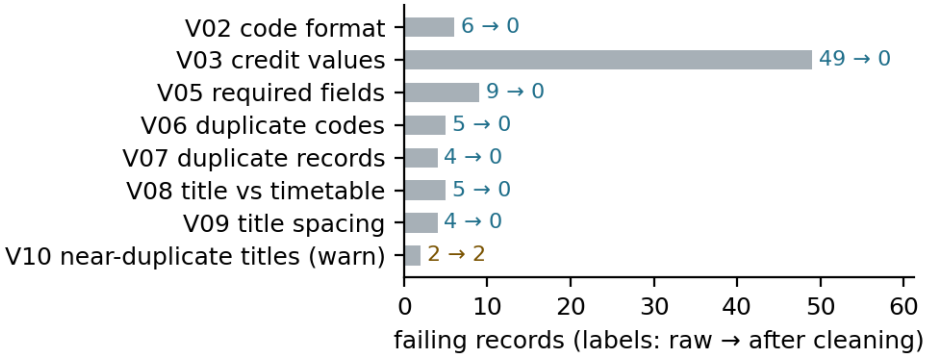


Figure 1: Failing records per check before and after cleaning [E28]. V10 flags two different courses with near-identical titles; they are kept on purpose. Full table: Table 1.

Corrupted copy. A DVC stage injects 15 faults with seed 209, among them a prerequisite cycle, an unknown prerequisite, total credits changed from 82 to 80, a below-600 cap breach and train/test leakage; all are caught [E29, E32, E30]. One planned fault, a per-semester credit cap, was not injected because no cap is stated [E31].

What people caught. While labelling the first item, hem436 saw “Other baskets (): 0 credits earned” for an MTech-AI student: the engine read other baskets from the student’s own one-stream programme. The fix follows the CSE structure and changed 100 of 600 gold answers, all MTech-AI [E33]. Later hem436 noticed answers telling M.Tech students they lacked IITGN UG courses; D5 removed these warnings and changed 53 gold answers and 0 query texts [E34]. Both have regression tests and came before any finished annotation. The AI fix would have changed 12 of the 180 students, so generation keeps the old basket view while gold answers use the fixed rule [E35, E36].

4 Part C: Labelling and agreement

Two annotators audited LLM paraphrases against the rule-engine answer in two rounds of 40 items. Agreement was low in round 1, we rewrote the guidelines, and it rose in round 2. Every disagreement was settled by the two annotators.

2 × 40	0.38	0.35	55 / 21 / 4	76
items, both annotators [E37]	decision kappa in round 2, up from 0.23 [E41]	category kappa in round 2, up from 0.17 [E41]	accept / edit / reject [E53]	gold test items [E8]

Task. Each item shows the student, the template query, the paraphrase, the gold answer and the rewritten justification. Annotators fill three levels in a fixed order: decision (accept, edit, reject), one error category, then the edit text (Appendix C). Deciding first keeps the category list from steering the decision. A1 (SMH) and A2 (MSB) labelled the same items in Label Studio 1.23.2, two disjoint rounds drawn from the test split with seed 208 [E37, E38, E39]. Round 1 used guidelines v1; round 2 used v2, which adds seven rules, each tied to a round-1 disagreement [E40].

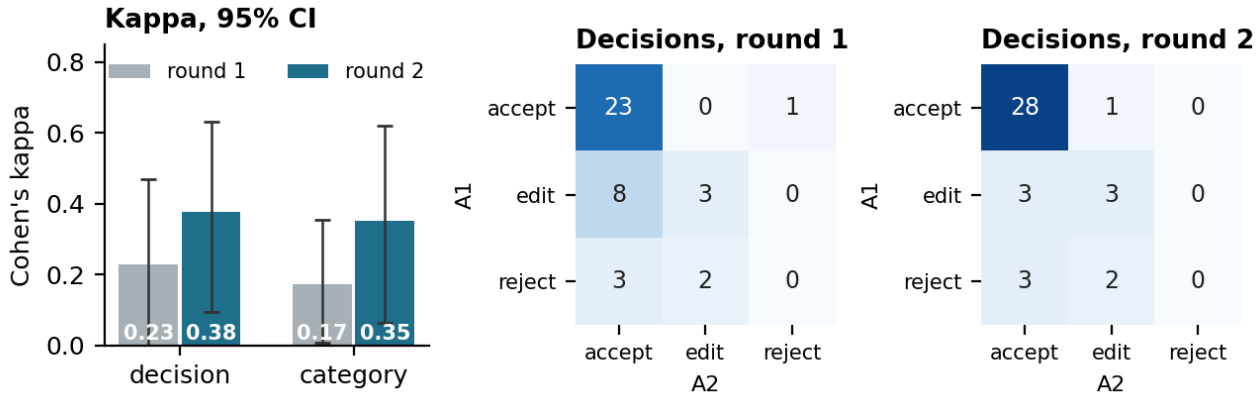


Figure 2: Left: Cohen's kappa with bootstrap 95% CI [E41]. Middle and right: decision confusion, A1 rows and A2 columns, round 1 and round 2 [E43].

Agreement. Decision kappa went from 0.227 [0.000, 0.469] to 0.377 [0.094, 0.630], and category kappa from 0.171 [0.007, 0.353] to 0.350 [0.064, 0.620], with 2000 bootstrap resamples [E41, E42]. Raw agreement went from 65.0% to 77.5% on decision and from 62.5% to 75.0% on category; disagreements fell from 17 to 12 [E44, E45]. The CIs overlap and the rounds use different items, so the gain is in the expected direction but not conclusive.

Why they disagreed. Round 1 was about strictness, not category confusion: A2 accepted 34 of 40 items and A1 24, and in 11 of the 17 disagreements A1 flagged an item that A2 accepted [E46, E47]. Causes: vague "almost there" justifications, wrong numbers in fluent text, the edit-or-reject threshold, and fixing correct gold from memory. In round 2 the main split left was wrong numbers in fluent text, and the paraphraser kept turning "coursework track" into "workshop track".

Adjudication. 50 items agreed fully. The other 30 (17 in round 1, 13 in round 2) are the 29 disagreements plus one item where both chose edit with different texts [E48]. The assistant drafted a proposal for each of the 30 with a one-line reason [E49]. A1 and A2 reviewed the 29 disagreements together: 28 approved, 1 overridden [E50]; A1 decided the extra item [E51]. None stayed unresolved [E52]. The gold test set is the 76 accepted or edited items (27 eligibility, 27 elective recommendation, 22 remaining requirements), frozen with sha256 `cd2e08dc...` [E54, E55].

Real form questions. 19 consenting responses were imported, 0 excluded; 8 came from form v1 and 11 from v2, which added three example questions after respondents were unsure what to ask [E9, E56, E57]. 9 questions were asked in person by classmates and entered by hem436 after rewording with ChatGPT; the 10 self-submitted rows keep the students' own words [E58]. Real questions are informal ("sem 3 of m.tech EE,SPML stream") and often fall outside the three intents. By kind: 9 remaining requirements, 5 elective advice, 2 eligibility, 3 other [E59].

5 Part D: Augmentation and active learning

Augmented pairs are kept only if the rule engine still gives the right answer; LLM paraphrases fail this far more often than the mechanical augmentations. Ranking by disagreement finds base-model failures faster than random, but plain top-k sees one intent.

1526	36 of 120	232 of 336	0.43
of 1610 augmented pairs kept [E6]	LLM paraphrases kept [E60]	base-model answers wrong [E63]	failures found by top-k at budget 100; random 0.30 [E65]

Checks. Typo and transcript children must leave the gold answer unchanged; transcript-length children must get the correctly recomputed answer. LLM paraphrase children must also keep every fact the question states (programme, stream, track, semesters completed, number of electives, term) and must not turn the question back on the student (guideline R4) [E61].

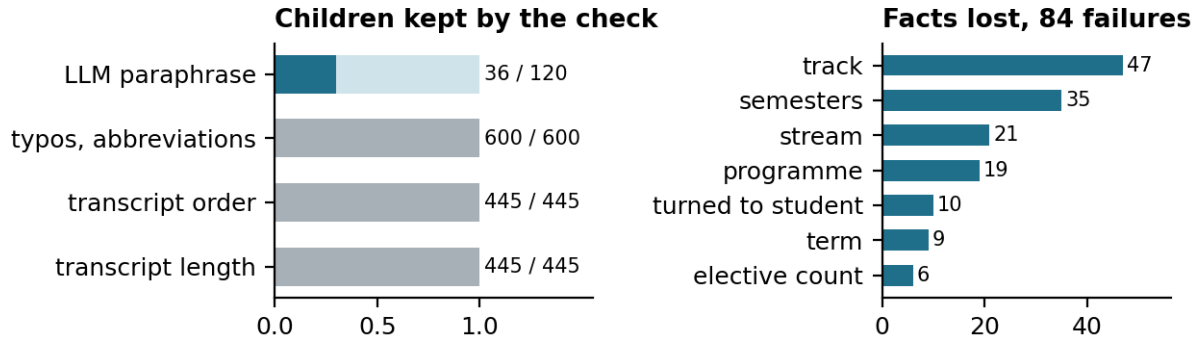


Figure 3: Left: share of children each check keeps [E60]. Right: facts lost by the failed LLM paraphrases (one paraphrase can lose several) [E62]. Full table: Table 2.

Most LLM paraphrases drop or change a fact: the track is lost in 47 failures (26 of them track alone), semesters completed in 35, stream in 21, and 10 turn the question to the student [E62]. This matches the “workshop track” errors the annotators caught. The typo and transcript checks pass at 1.0 by construction, since those transforms never touch the facts the engine reads.

Active learning. The base model (same Qwen, greedy) answered the 336 train pairs; 232 answers are wrong: wrong remaining 94, ineligible recommendation 64, wrong eligibility 56, invalid JSON 18, hallucinated code 2 [E63, E64]. Random is the mean of 5 seeds [E66]. The score and the failure test are the same comparison, so every top-k pick is a failure (precision 1 by construction) and the informative part is which failures each strategy reaches [E67]. Plain top-k takes every pick from one intent and covers 0.60 of the failure types; the intent-stratified variant, added after the first run, keeps most of the gain (0.38 at budget 100) and covers all failure types and intents [E65, E68]. This is a label-efficiency comparison, not a learning curve: no model is retrained.

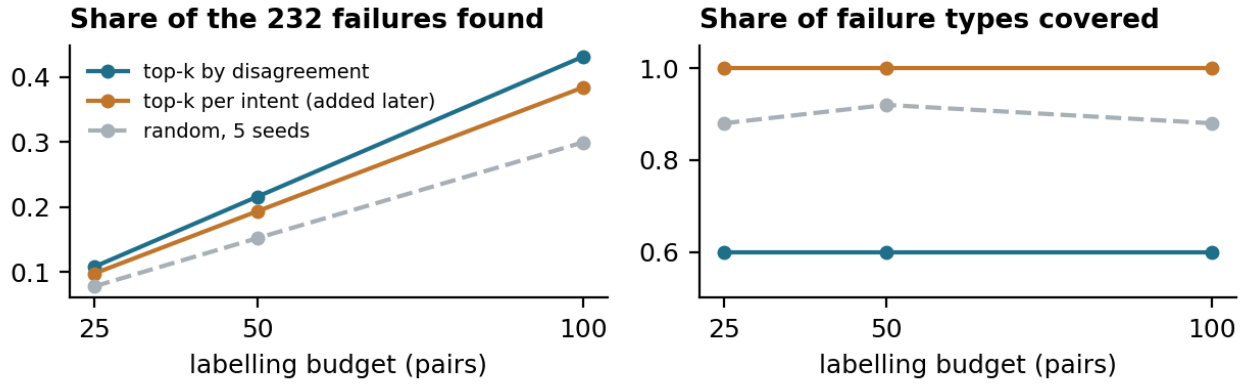


Figure 4: Left: share of the 232 base-model failures each strategy finds; right: share of failure types it reaches [E65]. Top-k finds failures fastest but only of one kind. Full table: Table 3.

6 Part E: Versioning and splits

Data and pipeline are versioned with DVC, and the splits keep students, wording and one whole programme apart so that test scores measure generalisation.

26	0	7 of 22	24	19
DVC pipeline stages [E69]	students shared between splits [E77]	templates used only in test [E76]	ICDT students held out [E78]	real questions in the shifted split [E9]

DVC. Human and raw inputs (sources, plans, form CSV, LLM cache, labelling files) are added with `dvc add`; everything else comes from the 26 stages, EE fetching frozen [E70, E69]. The remote is a local folder next to the repo [E71]. `dvc.lock` pins every output, and a full offline `dvc repro` reproduced courses, rules, students, pairs, paraphrases, splits, labels, the gold set and base-model answers byte for byte; only the run-time column of the metadata file differs [E72, E73]. The tag `a1-final` is set on the commit that contains this report [E74].

Splits. Students are assigned first, per programme (seed 207; 0.65 / 0.15 / 0.20), and each pair is then written with a template allowed in its split [E75]. All MTech-ICDT students form a programme holdout: it is the smallest programme and the only one without streams. Augmented pairs inherit their parent's split. All overlap checks are zero and no pair was dropped [E77]. Eligibility is balanced in every split (for example 17 eligible and 17 not in test) [E78].

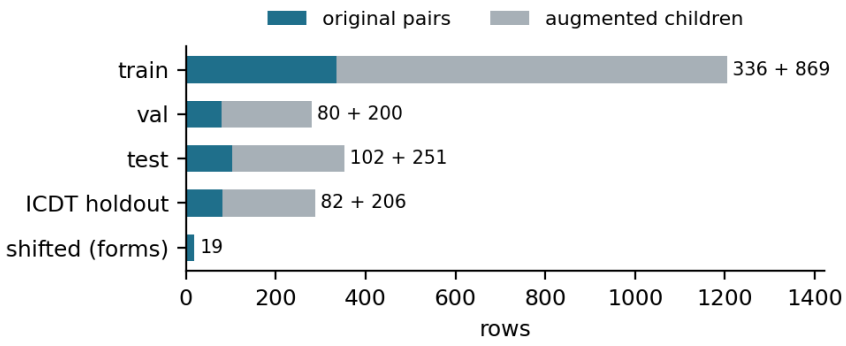


Figure 5: Rows per split: original pairs plus augmented children [E78, E9]. Full table with students and balance: Table 4.

7 Part F: Report and reproducibility

The repository documents the data, machine and labelling, and rebuilds everything with four commands.

252	8	3	3.10.12
tests pass [E79]	seeds in params.yaml [E80]	LLM backends, recorded per row [E84]	Python; DVC 3.67.1 [E81]

DATASET_CARD.md covers sources, sizes, balance, biases and intended use. ENVIRONMENT.md records the laptop (Ubuntu 22.04.5, Ryzen 5 2500U, CPU only), Python, DVC, Label Studio, the model digest and the Kaggle GPU (Tesla T4, CUDA 12.8, torch 2.11.0+cu128, transformers 5.16.1) [E81, E82]. The pipeline prompt README lists model, versions and backends per prompt [E83]. LABELLING_PROTOCOL.md records who labelled what, guidelines, item order, what annotators saw and a timeline matched to the exports. Rebuild:

```
python3.10 -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/dvc pull && .venv/bin/dvc repro && .venv/bin/pytest -q
```

LLM answers are replayed from the DVC-tracked cache, so a rerun gives the same text without a GPU.

8 Limitations

- Students are synthetic and valid by construction; no source states an enforceable PG prerequisite, so eligibility mostly tests offering and already-taken rules. Kappa rests on 40 items per round, so the CIs are wide.
- LLM backends are mixed: Ollama 0.1.38 could not load the qwen2 tokenizer, so the same weights ran on llama.cpp, and part of the paraphrases ran on a Kaggle T4. Each row records its backend; outputs differ across backends [E84, E85].
- Form examples were added mid-collection, which may pull later wording toward them. We have 19 real questions, below the target of 30; 9 were reworded with ChatGPT, so only 10 keep the students' own words [E86, E58].
- The EE site marks the ICDT list "to be updated soon", so ICDT answers rest on a partial list [E87].

Contributions

member	commits	work
Myank Singh Beniwal (MSB)	3	First data-loading commit; annotator A2 in both rounds (exports committed from his own account); joint adjudication with A1 (all 29 reviewed decisions are A1+A2). The docs record form distribution by hem436 only [E88, E89, E90].
Hemant (SMH, hem436)	45	Everything else: scope decisions, pipeline and rule engine (built with Claude Code), sources, form creation and sharing, annotator A1, joint adjudication, DVC, documentation and this report.

A Evidence index

Every number in the body, with the file that backs it (paths from the repo root) and the field or row.

id	what	file and field
Overview		
E1	4 programmes, thesis and coursework tracks	`docs/DECISIONS.md`, rows D1, D2, D3
E2	104 courses	`data/processed/courses.json`, list length; `supporting_files/B_engine/prereq_graph_report.json` `nodes`
E3	77 rules; 20 EE (`ee.*`); 1 not stated (max credits per semester)	`data/processed/rules.json`, `rules` keys, prefix `ee.`, `not_stated: true`
E4	180 students; CSE 64, EE 51, AI 41, ICDT 24; 92 thesis, 88 coursework; 1422 entries; 0 violations	`supporting_files/B_engine/transcripts_summary.json`, `n_students`, `programmes`, `tracks`, `transcript_entries`, `violations`
E5	600 pairs, 200 per intent	`supporting_files/B_engine/queries_summary.json`, `n_pairs`, `per_intent`
E6	1526 children kept of 1610	`supporting_files/E_splits/split_summary.json` `n_children_in` (sum of `child_rows_per_split`); `supporting_files/D_augmentation/verify_report.json` `children_in`
E7	80 audit items (40 per round)	`data/labels/audited.jsonl`, row count by `round`
E8	76 gold test items	`supporting_files/E_splits/gold_summary.json`, `n_gold`
E9	19 real form questions	`supporting_files/C_labelling/forms_import_log.json` `rows_kept`; `supporting_files/E_splits/shifted_summary.json` `n_rows`
Part A: Collection		
E10	8 advisories, CSE structure, timetable, 4 EE pages, each with sha256	`data/raw/sources/source_metadata.json` `sources[].sha256`; `data/metadata.csv` rows with `artifact` under `data/raw/sources/`
E11	3083 metadata rows (578 raw, 2505 processed)	`data/metadata.csv`, row count and `layer` counts
E12	180 / 600 / 1526 / 19 / 76 rows for students, pairs, children, forms, rules (76 of 77)	`data/metadata.csv`, `sample_id` prefixes `student`, `pair`, `aug`, `form`, `rule`
E13	50 CS/ES codes: 15 downloaded, 12 login, 23 no link	`supporting_files/A_collection/course_plan_fetch_log.csv`, last `outcome` per `code`
E14	25 EE codes: 7, 4, 14	`supporting_files/A_collection/ee_course_plan_fetch_log.csv`, same
E15	22 plan files (15 + 7)	`data/raw/course_plans/`, file count
E16	136 raw catalog records	`data/metadata.csv` rows with `artifact` = `data/raw/catalog_raw.json`; `supporting_files/A_collection/collection_log.md` section 7
E17	67 / 15 / 6 / 2 / 6 / 8 courses by category	`data/processed/courses.json`, `category`
E18	78 cleaning changes	`supporting_files/A_collection/cleaning_changes.csv`, data rows
E19	103 none stated, 1 manual; 0 enforceable	`data/processed/courses.json`, `prereq_source` and empty `prerequisites`
E20	CS 613 manual decision	`data/raw/prereq_manual.csv`, `decided_by`
E21	3 courses with recommended background	`data/processed/courses.json`, non-empty `recommended_background`
E22	104 nodes, 9 recommended edges, 0 missing, 0 cycles	`supporting_files/B_engine/prereq_graph_report.json`, `nodes`, `edges_by_type`, `missing_prereq_codes`, `cycles`
E23	decisions D4, D5	`docs/DECISIONS.md`, rows D4, D5
E24	seed 203	`params.yaml`, `seeds.transcripts`
Part B: Validation		
E25	25 checks, V01 to V25	`supporting_files/B_validation/processed/results.csv`, 25 rows; libraries in `requirements.txt`
E26	raw: 10 checks, 7 fail, 2 pass, 1 warn	`supporting_files/B_validation/raw/results.csv`, `status`

id	what	file and field
E27	processed: 24 pass, 1 warn (V10, 2)	`supporting_files/B_validation/processed/results.csv`, `status`, `n_failures`
E28	Figure 1 and Table 1 values	`supporting_files/B_validation/before_after.md`, table rows
E29	15 of 15 faults caught	`supporting_files/B_validation/corrupted/faults_caught.json`, `caught`
E30	total credits 82 to 80	same, F05 `description`
E31	1 fault not injected	`data/interim/corrupted/faults.json` (output of DVC stage `validate_corrupted`), `not_injected`
E32	seed 209	`params.yaml`, `seeds.validation_corruption`
E33	100 of 600 gold answers changed (AI fix)	`supporting_files/B_engine/gold_fix_mtech_ai_baskets.md`, "Effect"
E34	53 gold answers, 0 query texts changed (D5)	`supporting_files/B_engine/gold_fix_ug_background.md`, "Effect"
E35	12 of 180 students the AI fix would change	`supporting_files/B_engine/gold_fix_mtech_ai_baskets.md`, last section
E36	old basket view kept for generation	`src/engine/rule_engine.py`, `role_of` docstring
Part C: Labelling and agreement		
E37	40 items per round, 2 rounds	`supporting_files/C_labelling/round1/agreement_summary.json` and `round2/`, `n_items_compared`
E38	seed 208	`params.yaml`, `seeds.annotation_order`
E39	Label Studio 1.23.2	`ENVIRONMENT.md`, Python and tools table
E40	seven v2 rules (R1 to R7)	`supporting_files/C_labelling/round1/disagreement_analysis.md`, "Changes for guidelines v2"
E41	kappa 0.227, 0.171, 0.377, 0.350 and their 95% CIs	`supporting_files/C_labelling/kappa_ci.json`, `round*_decision`, `round*_error_category`
E42	2000 resamples	same, `n_boot`
E43	confusion matrices	`supporting_files/C_labelling/round1/confusion_decision.csv` and `round2/`
E44	raw agreement 65.0 / 62.5 and 77.5 / 75.0%	`supporting_files/C_labelling/round1/agreement_summary.json` and `round2/`, `percent_agreement`
E45	17 and 12 disagreements	same, `n_disagreements`
E46	A2 accepted 34, A1 24 (round 1)	`supporting_files/C_labelling/round1/agreement_summary.json`, `label_counts`
E47	11 of 17: A1 flagged, A2 accepted	`supporting_files/C_labelling/round1/disagreement_analysis.md`, "The main pattern"
E48	50 agreed; 17 + 13 adjudicated	`supporting_files/C_labelling/adjudication_log.json`, `agreed`, `adjudicated`
E49	30 proposals (29 + R2-Q0196)	`supporting_files/C_labelling/adjudication_proposals.json`, keys other than `_about`
E50	28 approved, 1 overridden, all A1+A2	`supporting_files/C_labelling/adjudication_decisions_export.json`, `choice`, `decided_by`
E51	extra item decided by A1	`data/labels/audited.jsonl`, `decided_by` of R2-Q0196
E52	0 unresolved	`supporting_files/C_labelling/adjudication_log.json`, `unresolved_items`
E53	55 accept, 21 edit, 4 reject	`data/labels/audited.jsonl`, `final_decision`
E54	gold by intent 27 / 27 / 22	`supporting_files/E_splits/gold_summary.json`, `by_intent`
E55	sha256 `cd2e08dcb56f5716b06981e096c3c93fd417abab1989b5d22d45befeafbc383f`	`data/gold/GOLD_SHA256.txt`, line 1
E56	0 excluded	`supporting_files/C_labelling/forms_import_log.json`, `rows_excluded`
E57	8 v1, 11 v2	`supporting_files/E_splits/shifted_summary.json`, `by_form_version`
E58	10 self-submitted, 9 entered after ChatGPT rewording	same, `by_provenance`; `supporting_files/C_labelling/forms_log.md`
E59	9 / 5 / 2 / 3 by kind	same, `by_question_kind`
Part D: Augmentation and active learning		

id	what	file and field
E60	36 of 120 paraphrases; Figure 3 left and Table 2	`supporting_files/D_augmentation/invariance.csv`, all rows
E61	facts the paraphrase check compares	`src/augment/verify.py`, `slot_issues`
E62	track 47 (26 alone), semesters 35, stream 21, turned to student 10	`supporting_files/D_augmentation/invariance.csv`, `fail_reasons`, counts summed over every reason that names the slot
E63	336 pool pairs, 232 failures	`supporting_files/D_active_learning/scores_summary.json`, `n`, `failed`
E64	94 / 64 / 56 / 18 / 2 by failure type	same, `failure_types`
E65	0.43, 0.38, 0.30 at budget 100; 0.60 types; Figure 4 and Table 3	`supporting_files/D_active_learning/label_efficiency.csv`, `failures_captured`, `failure_types_covered`, `intents_covered`
E66	5 random seeds	`params.yaml`, `seeds.active_learning`
E67	precision 1 by construction; not a learning curve	`supporting_files/D_active_learning/summary.md`, "Reading the first column"
E68	stratified variant added later	same, paragraph after it
Part E: Versioning and splits		
E69	26 stages, EE fetch frozen	`dvc.yaml`, `stages`, `fetch_ee_pages: frozen`
E70	inputs added with dvc add	`data/raw/*.dvc`, `data/raw/forms/responses_raw.csv.dvc`, `data/interim/llm_cache.jsonl.dvc`, `data/labels/round*/*.dvc`
E71	local remote `../dvc_remote`	`.dvc/config` `remote "localremote"`; `README.md`
E72	every output pinned	`dvc.lock`, stage `outs` hashes
E73	byte-for-byte rerun	`supporting_files/F_reproducibility/repro_check.txt`, `sha256sum -c` lines, `dvc status`
E74	tag a1-final	`HANDOFF_A2.md`, first paragraph
E75	seed 207; 0.65 / 0.15 / 0.20	`supporting_files/E_splits/split_summary.json`, `seed`, `fractions`
E76	7 of 22 templates test-only	same, `held_out_templates`; `supporting_files/B_engine/queries_summary.json` `per_template`
E77	all overlaps 0, 0 dropped	`supporting_files/E_splits/split_summary.json`, `overlap_checks`, `all_overlaps_zero`, `n_dropped_pairs`
E78	Figure 5 and Table 4 values; 24 ICDT students; 17 / 17 in test	same, `student_assignment`, `original_rows_per_split`, `child_rows_per_split`, `rows_per_split`, `eligibility_by_split`, `original_rows_by_programme`
Part F, limitations, contributions		
E79	252 tests pass	`supporting_files/F_reproducibility/pytest.txt`, last line
E80	8 seeds	`params.yaml`, `seeds`
E81	Ubuntu 22.04.5, Ryzen 5 2500U, Python 3.10.12, DVC 3.67.1	`ENVIRONMENT.md`, Machine and Python and tools tables
E82	Tesla T4, CUDA 12.8, torch 2.11.0+cu128, transformers 5.16.1	`ENVIRONMENT.md`, GPU row
E83	model, versions, backends per prompt	`prompts/pipeline/README.md`, Model and Prompts
E84	3 backends; Ollama 0.1.38, llama.cpp, Kaggle T4	`supporting_files/D_augmentation/llm_backend.md`, sections 1, 2, 4
E85	backend per row	`data/processed/queries_para.jsonl`, `para_backend`
E86	target of 30 real queries	assignment brief (not in the zip), Section 8, Project 4: "at least 30 real queries"
E87	ICDT list "to be updated soon"	`supporting_files/A_collection/collection_log.md` section 7; `DATASET_CARD.md`
E88	45 / 3 commits	git history: `git log --format=%an | sort | uniq -c`
E89	A2 exports committed by MSB	git commits f395c8a, 709292f; `LABELLING_PROTOCOL.md` section 9
E90	form shared by hem436	`supporting_files/C_labelling/forms_log.md`, row 2
E91	session logs cover prompts 1 to 61	`prompts/assistant/claude_code_session_2026-10-18.md`, last heading "Prompt 61"

B AI assistance

What the assistant did. Claude Code (model claude-opus-5-5) wrote most of the code through sub-agents whose prompts are in `prompts/assistant/`: `agent_common_rules.md`, `agent_A_curriculum.md`, `agent_A2_ee_pages.md`, `agent_B_engine.md`, `agent_B2_engine_revision.md`, `agent_C_labelling.md`, `agent_D_validation.md`, `agent_E_llm_augment_al.md`, `agent_F_splits_dvc_docs.md`, `agent_F2_resplit.md`, `agent_G_report.md` and `agent_G2_report_redesign.md` (this report) and `agent_H_mock_grading.md` (mock grading). The user's own prompts are logged in `claude_code_session_2026-10-17.md` and `claude_code_session_2026-10-18.md`, which cover prompts 1 to 61 [E91]. The assistant also drafted the 30 adjudication proposals (people made every final call), built the labelling kit pages and the adjudication review page, and built and edited the Google Form in the browser at the user's request.

ChatGPT. hem436 used ChatGPT to reword 9 questions that classmates asked in person before entering them in the form (`prompts/assistant/chatgpt_form_rewording.md`; the prompt and the original wording were not kept).

Pipeline LLM. Qwen2.5-1.5B-Instruct (Q4_K_M via llama.cpp, or fp16 via transformers on Kaggle) paraphrases queries and justifications and gives base-model answers. Its prompts are files in `prompts/pipeline/`: `paraphrase_query_v1.txt`, `paraphrase_justification_v1.txt`, `augment_paraphrase_v1.txt` and `base_model_answer_v1.txt`. Model, digest and backend are stored with every output, and the README there lists model, versions and backends per prompt and the Qwen default system prompt.

What people did. All labelling (A1 and A2, both rounds), every adjudication decision, the scope and modelling decisions in `docs/DECISIONS.md`, both gold-bug findings, and form sharing and distribution. The assistant declined to make commits under Myank's identity; his commits are his own.

C Guidelines v2 and label levels

Label levels, in this order: 1) decision: accept, edit or reject; 2) one error category, by priority: hallucinated course, wrong fact or policy, meaning drift, incomplete justification, ambiguous query, fluency (or none); 3) edit text, only for edit. The label set and the Label Studio config did not change between rounds.

Changes from v1 to v2, each tied to round-1 disagreements:

- R1: a justification must state the gold outcome and every number, course and open item; pure encouragement is an incomplete justification.
- R2: any number or status that contradicts gold or is not in gold is a wrong fact, including vague text that implies a wrong status.
- R3: edit when the fix touches at most two sentences or one query phrase, otherwise reject; a missing fact alone is never a reason to reject.
- R4: added scope, "available" for "remaining", or a question turned back on the student is meaning drift.
- R5: with two problems, pick the category by priority and fix both texts.
- R6: gold is the truth; do not correct it from memory, write "GOLD?" instead.
- R7: formatting and tone are not errors; edit or reject always needs a category other than none.

D Detailed tables

Table 1: Failing records before and after cleaning. Source: `supporting_files/B_validation/before_after.md`.

check	what it tests	raw	after
V02	code format	6	0
V03	credits are integers in range	49	0
V05	required fields present	9	0
V06	no duplicate codes	5	0
V07	no exact duplicate records	4	0
V08	title matches timetable	5	0
V09	title spacing and punctuation	4	0
V10	near-duplicate titles (warning)	2	2

Table 2: Augmentation checks. Source: `supporting_files/D_augmentation/invariance.csv`.

augmentation	check	children	kept	rate
LLM paraphrase	gold and stated facts unchanged	120	36	0.30
typos and abbreviations	gold unchanged	600	600	1.00
transcript order	gold unchanged	445	445	1.00
transcript length	gold recomputed correctly	445	445	1.00

Table 3: Share of the 232 failures captured per budget, and coverage. Random is the mean of 5 seeds. Source: `supporting_files/D_active_learning/label_efficiency.csv`.

strategy	25	50	100	types	intents
top-k by disagreement	0.11	0.22	0.43	0.60	1
stratified by intent (added later)	0.10	0.19	0.38	1.00	3
random	0.08	0.15	0.30	0.88 to 0.92	3

Table 4: Split sizes. Students are those assigned; pairs per programme are CSE / AI / EE / ICDT. Source: `supporting_files/E_splits/split_summary.json`; shifted row from `supporting_files/E_splits/shifted_summary.json`.

split	students	pairs	children	rows	eligible / not	by programme
train	101	336	869	1205	56 / 56	127 / 90 / 119 / 0
val	24	80	200	280	13 / 13	35 / 21 / 24 / 0
test	31	102	251	353	17 / 17	49 / 21 / 32 / 0
ICDT holdout	24	82	206	288	14 / 14	0 / 0 / 0 / 82
shifted (forms)	n/a	19	0	19	n/a	n/a